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Solar Irradiance Forecasting

Addressing Train-Serve Skew in Solar Irradiance Forecasting in Northern Vietnam: A Robustness and Calibration Framework

Students:

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ONE LOVE. ONE FUTURE.

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Contribution table

Name	ID	Contribution
Nguyễn Thành Nam	20233870	• Literature Review & Reference Retrieval • Baseline Model Development & Optimization • Construction of Monte Carlo Perturbation Framework - Phase 1 • Report + Slide + Presentation
Nguyễn Hương Thảo	20233879	• Literature Review & Reference Retrieval • Data Collection & Data Cleaning • Exploratory Data Analysis (EDA) and Feature Engineering • Results Synthesis & Evaluation • Report + Slide + Presentation
Nguyễn Thị Anh Thơ	20233880	• Literature Review & Reference Retrieval • Exploratory Data Analysis (EDA) • Implementation of Conformal Prediction Algorithm - Phase 2 • Evaluation of Phase 2 Results • Report

1. Introduction

Motivation:

- Global horrizontal irradiance **(GHI)** directly determines the power output of photovoltaic systems
- Northern Vietnam: a key renewable energy hub facing power shortage risks
- Highly complex microclimate: 4 distinct seasons · urban fog · sudden storms

Reseach Gap:

- Train-Serve Skew: models trained on perfect historical data but must rely on imperfect weather forecasts during real-world operation
- Point forecasting completely hides operational risks for grid operators

This study

- Evaluate model robustness under NWP forecast errors via Monte Carlo perturbation
- Construct prediction intervals integrating both input uncertainty + model uncertainty via Conformal Prediction
- Empirically validated across 3 microclimates: Hanoi · Bac Giang · Son La

2. Related work

[1] Comparison of forecasting performance between the Clearness Index (Kt) and the Clear Sky Index (Kc) approaches:

- ANN with Kt time series outperform linear model constructed using (Kc)

[2] Comparison of GHI forecasting using Kt and Kc across two regions in Algeria:

- The models achieve higher accuracy in regions with stable climatic conditions
- The Kt more accurate performance compared to the Kc in the context of this study

[3] Combination of K-means clustering with Support Vector Regression (SVR) to predict Kt:

- Forecasting the Kt instead of predicting GHI directly is employed to improve data stationarity and reduce the impact of periodic seasonal fluctuations.

[1] P. Lauret, R. Alonso-Suárez, J. Le Gal La Salle, and M. David, “Solar forecasts based on the clear sky index or the clearness index: Which is better?” Solar, vol. 2, no. 4, pp. 432–444, 2022. [Online]. Available: https://www.mdpi.com/2673-9941/2/4/26

[2] C. Souad, N. Tarek, N. Omar, and Y. Farid, “Predicting the clear-sky and learness indices for ghi estimation under two algerian climate,” 2025 2nd International conference on Advances in Electronics, Control and Communication Systems (ICAECCS), pp. 1–4, 2025. [Online]. Available: https://api.semanticscholar.org/CorpusID:285803141

[3] Z. Yang, Y. Li, P. Shen, y. Huang, K. Zhang, and S. Li, “Research on clearness index support vector regression method for solar radiation forecasting,” 12 2021, pp. 69–74.

2. Related work

[4] G. Vishnu, D. Kaliyaperumal, and S. R. Salkuti, “Seasonal solar irradiance forecasting using artificial intelligence techniques with uncertainty analysis,” Scientific Reports, vol. 14, 08 2024.

[4] A comparative point forecasting approach using SARIMAX, SVR, and LSTM models to forecaste directly GHI and interval forecasting that uses a parametric Laplace distribution and construct confidence intervals

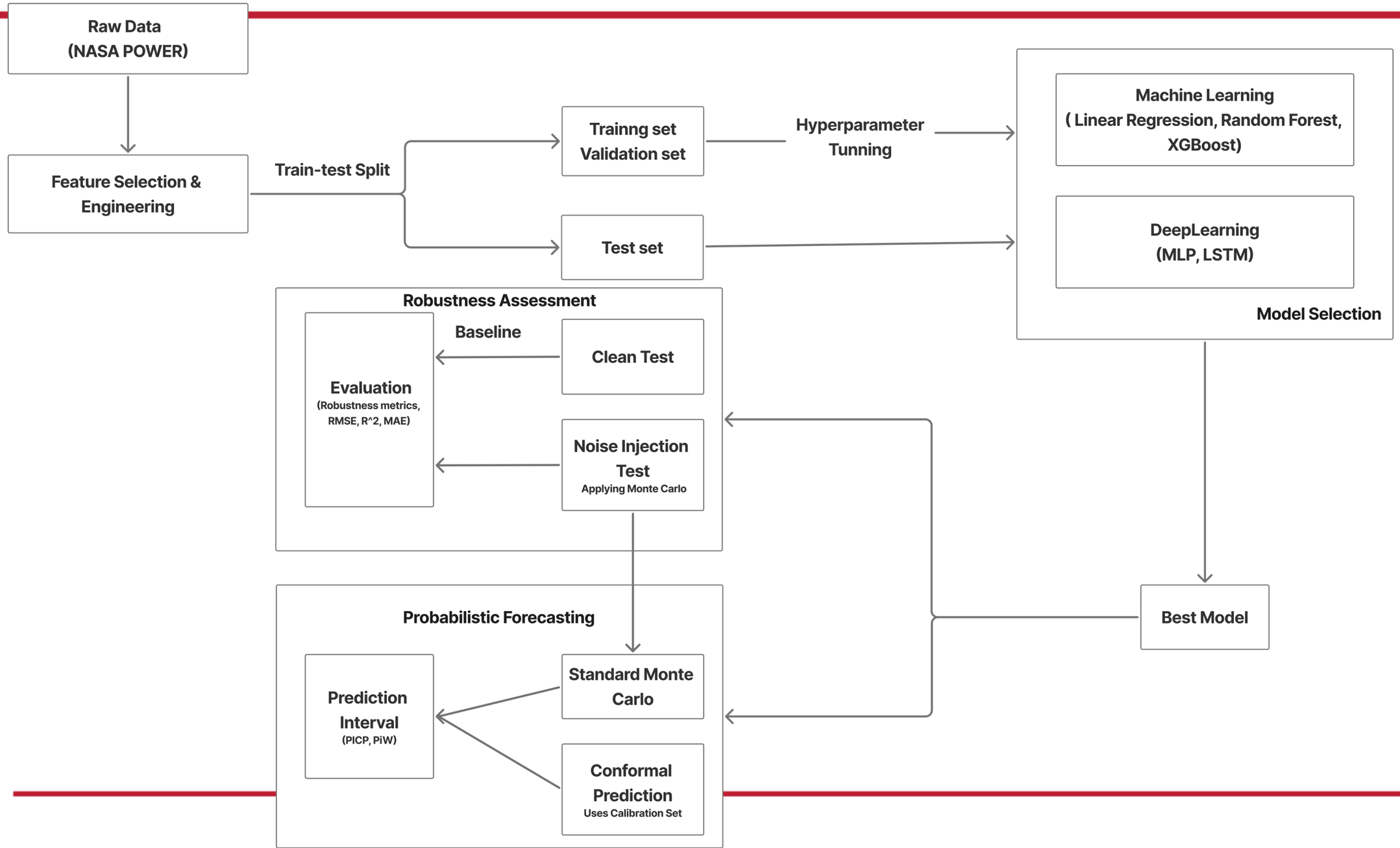
- SVR is best-performing model overall.
- The Coverage Rate of SVR base on Laplace distribution,
- The model achieves high coverage rates across all confidence levels, indicating reliable uncertainty estimation

Seasons	Coverage rate			
	CI: 95%	CI: 90%	CI: 85%	CI: 80%
Winter	93	90.3	89.7	89
Summer	98.4	97.9	97	95
Monsoon	98.6	95	93.4	90.9
Post-monsoon	97.6	95	91.6	88

2. Related work

Ref.	Methodology	Target	Addressed Weather Uncertainty?	Limitations / Gaps
[1]	Benchmarking ML models	K_t	No	Assumes perfect weather inputs.
[2]	ARIMA Time-Series	K_t, Kc	No	Fails in highly unstable climates.
[3]	K-means + SVR	K_t	No	Assumes perfect weather inputs.
[4]	SVR + Laplace Distribution	GHI	Partial (Reactive)	Analyzes residuals only; no proactive stress-testing.
Ours	**Monte Carlo + Conformal Prediction**	K_t	**Yes (Proactive)**	**Addresses Train-Serve Skew directly;** **Construct prediction intervals**

3. Dataset and Proposed method



Dataset

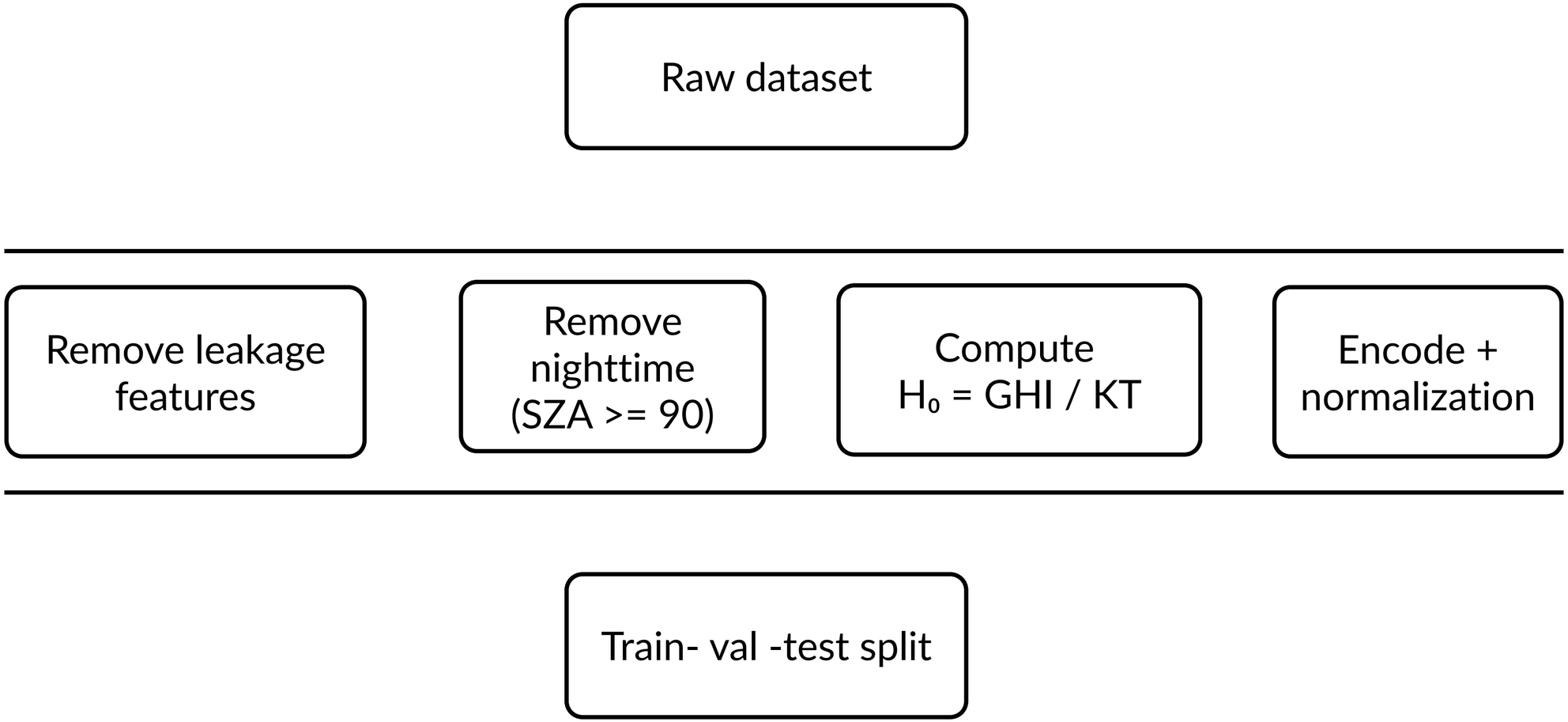
- Dataset from: NASA POWER
- Region: Hanoi (21.0285, 105.8542)
  Sơn La (21.327, 103.914)
  Bắc Giang(21.272, 106.194)
- Time period: 01/01/2020 - 31/12/2025
- Temporal resolution: 1-hour interval
- Number of column: 12 column (time, meteorology, astronomy, and solar radiation)

Features:

- **Time variables:** year, month, and hour.
- **Meteorological variables**: air temperature , relative humidity, corrected precipitation rate, and wind speed
- **Astronomical variables:** solar zenith angle (SZA).
- **Solar radiation variables:** direct normal irradiance (DNI), diffuse horizontal irradiance (DIFF), clearness index (KT), global horizontal irradiance (GHI), and clear-sky global horizontal irradiance (CLRSKY_GHI)

Feature engineering

- **Target transformation:** Forecast Clearness index (K_t) instead of GHI



Phase 1: Robustness Evaluation via Monte Carlo Perturbation

Best model for each region

Noise design:

- Perturbed variables (4 biến NWP): T2M, RH2M, WS10M, PRECTOTCORR

- Gaussian noise: $\sigma_i = \lambda * \ std(x_i)$
  3 noise level: Low ($\lambda = 0.25$), Medium ($\lambda = 0.5$), High ($\lambda = 1.00$)

- Perturbation probability:
  - T2M - 0.3
  - RH2M - 0.4
  - WS10M - 0.5
  - PRECTOTCORR - 0.8

Monte Carrlo Simulation

- Timestep-level: 50% test samples inject noise
- M = 50 independent trials / noise level

Evaluation

- Deterministic: RMSE_mean, MAE_mean, R^2_mean
- Robustness Score

Phase 2: Probabilistic Forecasting

Step 1 — Calibrate Local Model Uncertainty (LWCP)

- Collect residuals from the validation set:
  $\varepsilon_i = y_i - \hat{y}_i$
- For each test sample, find K = 25 nearest neighbours in the validation set conditioned on (hour_sin, hour_cos)
- Extract empirical quantiles from K-neighbour residuals

Step 2 — Integrate Input Uncertainty

- Expand the interval using σ̂_input (MC-derived std from Phase I)

Evaluation Metrics:

- PICP-P1: Coverage from MC intervals (Phase I) — baseline
- PICP-P2: Coverage after Conformal Prediction

- PIW: Average prediction interval width

5. Experimental Result

Performance of Base Models

- Hà Nội: MLP archieves highest performance (RMSE: 83.85, R-Squared: 0.87).
- Sơn La: LSTM archieves highest performance (RMSE: 73.30, R-Squared: 0.90).
- Bắc Giang: XGBoost archieves highest performance (RMSE: 77.85, R-Squared: 0.88).

=> No single algorithm that is optimal for every area; differences in terrain and microclimate determine which model is most suitable.

5. Experimental Result

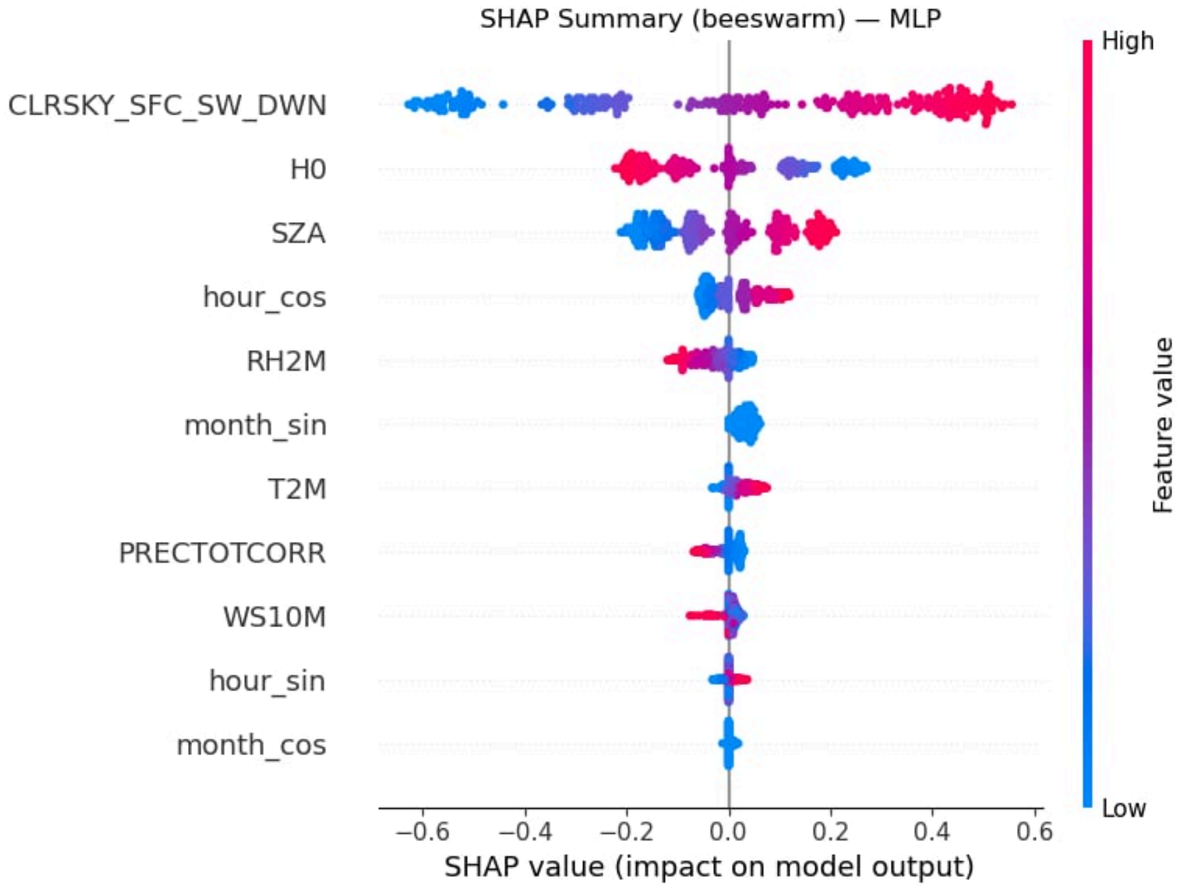
Phase I: Robustness Evaluation

		RMSE	MAE	R2	Robustness Score
Hà Nội	clean	83.85	569.00	87	100
	low	86.35	**58.73**	86	**97**
	medium	91.00	61.81	84	**91**
	high	100.69	67.63	80	80
Sơn La	clean	**73.50**	**48.06**	**90**	100
	low	**75.16**	50.59	**89**	97
	medium	**76.37**	**51.50**	**88**	96
	high	**80.40**	**54.28**	**87**	**90**
Bắc Giang	clean	77.85	56.38	88	100
	low	91.30	63.29	84	83
	medium	95.73	65.88	82	77
	high	102.18	69.75	79	69

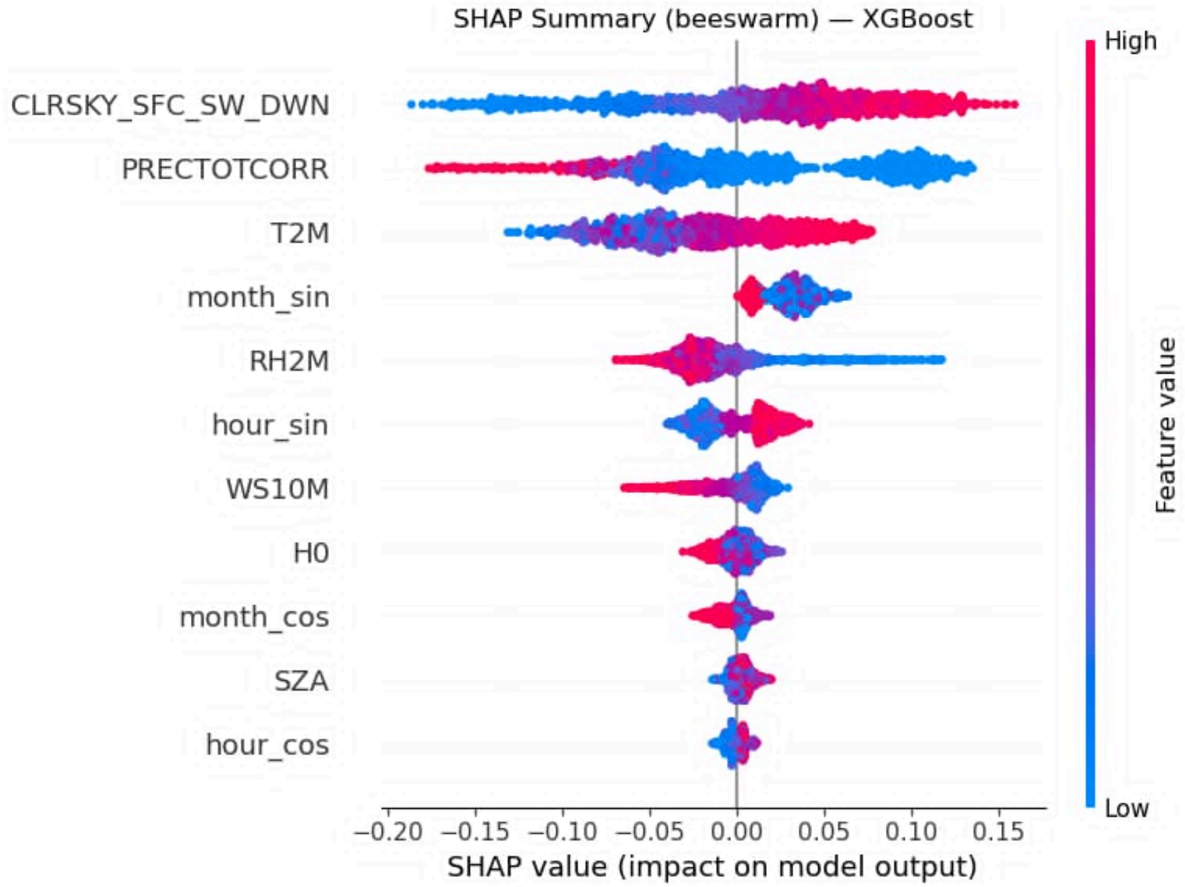
- Hanoi demonstrates the highest robustness
- Bac Giang is the most vulnerable region

5. Experimental Result

Phase I: Robustness Evaluation



Hanoi



Bac Giang

5. Experimental Result

Phase II: Probabilistic Forecasting

- Coverage of Phase 1 is extremely low (PICP-P1 = 7% for Sơn La)
- Coverage of Phase 2 (PICP - P2) exceeding 0.85 (85%) across all scenarios

		PICP - P1 (%)	PICP - P2 (%)	PIW - P2 (W/m²)
Hà Nội	clean	NaN	85.0	251
	low	23.2	89.6	282
	medium	38.0	92.0	308
	high	53.1	94.1	343
Sơn La	clean	NaN	85.1	204
	low	7.0	88.7	219
	medium	13.0	90.3	232
	high	23.9	92.4	254
Bắc Giang	clean	NaN	84.9	231
	low	34.1	94.8	296
	medium	42.0	95.8	312
	high	49.4	96.4	330

6. Conclusion and feature work

Conclusion:

- Proposed a two-phase framework addressing Train-Serve Skew in GHI forecasting
- Optimal model architecture depends on local microclimate: MLP (Hanoi) · LSTM (Son La) · XGBoost (Bac Giang)
- All models degrade under NWP noise; Bac Giang is the most vulnerable
- PICP-P1 extremely low (e.g. 7% in Son La) → input noise alone is insufficient
- Conformal Prediction restores reliability: PICP ≥ 85% across all scenarios
- Framework delivers quantifiable risk warnings for grid operators

Future work:

- Validate framework on real-world SCADA data from active solar plants
- Explore advanced CP variants to further narrow PIW for Northern Vietnam regions
- Integrate probabilistic forecasts into Battery Energy Storage System (BESS) optimization

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THANK YOU!

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